

ADITYA PAGARE

Applied AI Engineer | AI Systems Engineer | Solutions & Frameworks | Edge AI | Embedded AI

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PROFESSIONAL SUMMARY

Applied AI and embedded systems engineer with an Electronics and Telecommunication foundation. Core strength covers hardware registers, sensors, signal processing, embedded constraints, and software systems. Work centres on converting physical-world observations into requirements, experiments, and working AI systems. This profile combines hardware understanding, ML/LLM engineering, research methods, and customer context. Operating style is direct ownership in ambiguous problems, with domain study before proposing a solution.

CORE SKILLS

AI/ML Engineering: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP); PyTorch, TensorFlow Lite (TFLite), scikit-learn; model training, model evaluation, transfer learning, benchmarks, ablation studies, error analysis, reproducibility.

Generative AI & LLM Systems: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), embeddings, semantic search, vector databases; prompt design, tool/function calling, agent workflows (ReAct/BDI), structured outputs, LLM evaluation, safety checks.

Edge AI & Embedded Systems: Edge AI, on-device AI, on-device inference, model quantization, latency and memory profiling; embedded systems, embedded Linux, RTOS, real-time systems, microcontrollers; Android/JNI, hardware-aware deployment, sensor fusion, signal processing.

Languages & Systems Programming: Python, C/C++, Rust, Kotlin, TypeScript, SQL, Bash, Linux; systems programming, memory management, concurrency; cryptographic systems (AES-256-GCM, SHA-256/HMAC); Git, GitHub Actions, CI/CD.

AI Systems Architecture & Delivery: Systems design, AI pipeline design, API design, FastAPI, REST APIs; SQLite, vector retrieval (HNSW/FTS5); safety-critical design, performance tuning, technical documentation; customer discovery, solution scoping, prototyping, customer-facing delivery.

Research & Technical Leadership: Benchmark design, ablation studies, validation, technical writing, manuscript preparation; cross-functional ownership, research leadership, independent technical decisions, first-principles problem solving.

WORK EXPERIENCE

Applied AI Research Engineer, Nutan College of Engineering & Research

Pune, India | Crash-safety AI validation for insurance/fleet evidence review | NHTSA CISS validation | IJVSS journal-track manuscript

Jan 2026 – Jun 2026

- Defined VISTA applied-AI research requirements from sensor, forensic, insurance, and fleet inputs. Converted them into validation targets and manuscript-ready system requirements.
- Built the validation layer using CrashBench outputs, **100+ vehicle observations**, NHTSA CISS crash-pulse data, and simulations. Achieved **95.0% crash detection** and **11.98 km/h Delta-V MAE**.
- Designed a **tamper-evident evidence architecture** for inertial sensing, OBD-II, audio/visual checks, and **SHA-256/HMAC verification**. Made crash outputs reproducible and reviewable for insurance/fleet use.

Embedded Edge Systems Engineer, Bio-Inspired Robotics Research Group

Remote | Cross-functional rover systems engineering for Theo Jansen walking rover | edge control, telemetry, sensors, and voice-command safety constraints

Feb 2026

- Engineered and tested the rover control stack across Raspberry Pi edge control, Arduino motor/sensor loops, browser telemetry, and voice commands. Converted rough-terrain behaviour into engineering constraints.
- Set uncertainty limits, safety boundaries, and fail-safe rules with robotics, controls, and software teams. Kept motion, obstacle response, and operator commands predictable on uneven terrain.
- Validated field readiness on **5 surfaces**. Results: **sub-100 ms control latency**, **4.00 Hz telemetry**, **115–132 ms obstacle response**, **2.9% ultrasonic error**, and voice-parser lift from **88.93% to 95.61%**.

AI Systems Engineer, PageAI Systems

Pune, India | AI engineering practice for voice AI automation, client platforms, and microservice delivery

Feb 2023 – Nov 2025

- Built voice AI automation for **15+ client workflows**: real estate sales calls, lead qualification, property matching, CRM handoff, and escalation.
- Designed and deployed **cost-controlled LLM automation across 15+ client systems**. Standardised **6 integration layers**: voice/LLM wrappers, prompt routing, structured extraction, deterministic routing, dashboards, and REST/API handoffs.
- Delivered **15+ client-facing AI automation and web systems**. One example was a **6-microservice adaptive assessment platform** using FastAPI, Node.js, PostgreSQL, Redis, Docker, and Bayesian Knowledge Tracing.

PROJECTS

AURA — Private On-Device AI Agent Runtime

Rust, Android/Kotlin, local LLM inference, SQLite, vector search, full-text retrieval, JNI, agent workflows

Private alpha: architecture docs & demo available on request

Mar 2026 – Present

Headline metric: **116K LOC** and **2,051 tests**; Rust/Android AI runtime tested on phone-class hardware.

- Built an on-device AI agent for reminders, routines, messages, and app actions. Kept personal context local and targeted **under 500 MB RAM** on phone-class hardware.
- Implemented a deterministic-plus-LLM engine with **sub-10 ms matching** for repeat intents. Used LLM reasoning for new requests, typed action contracts, rollback paths, and side-effect isolation.
- Secured private memory with SQLite, vector/full-text retrieval, **AES-256-GCM/Argon2id vaulting**, consent gates, and notification safeguards.

BFRB Sense — Wearable Behaviour Recognition Edge AI

Python, PyTorch, Polars, SciPy, IMU/ToF/thermal, CNN-SE-GRU, 3D CNN, TFLite

Faculty-guided prototype • Child Mind Institute dataset • 0.93 internal vs 0.88 baseline

May 2025 – Sep 2025

- Led ML-system design for a wearable edge-AI prototype for early BFRB intervention. Converted multimodal sensor streams into **18-class behaviour recognition** on the Child Mind Institute dataset.
- Improved the data layer for physical signal validity: quaternion repair, handedness canonicalization, ToF symmetry, outlier filtering, **128-step windows**, and robust scaling. Score improved from **0.88 to 0.93**.
- Trained modality-aware ensembles with **IMU CNN-SE-GRU** branches, all-sensor fusion, ToF 3D CNN volumes, and AdamW/CUDA restarts. Prepared outputs for **TFLite edge inference**.

EDUCATION

Bachelor of Technology, Electronics & Telecommunication Engineering, PCET's Nutan College of Engineering & Research

Pune, India | Focus: **edge AI systems**, **embedded systems**, AI/ML frameworks, IoT, signal processing, sensors, microcontrollers, backend/API design, databases, and cloud deployment

Aug 2022 – Jun 2026

AWARDS AND LEADERSHIP

- Research leadership: First-author VISTA manuscript submitted to IJVSS. Converted crash-safety evidence needs into NHTSA CISS-validated results: 95.0% detection and 11.98 km/h Delta-V MAE.
- Product innovation: RANK 1 for an AI sign-language-to-speech prototype using computer vision, speech, and LLM-assisted interaction.
- Technical leadership: Nutan Incubation Centre participant, technical hackathon host, GitHub-visible open-source contributor, and demo/documentation owner.